

Combining Technical Indicators and Equal-Budget Metaheuristic Optimization for Short-Term Mini-Index Futures Prediction

Abstract

Short-horizon trend classification in futures markets is vulnerable to non-stationarity, weak signal-to-noise ratios, and temporal information leakage. An earlier conference precursor combined technical indicators, Information Gain feature selection, and genetic-algorithm tuning, but used randomized cross-validation, a single metaheuristic, and no economic assessment. This study advances that research line through an auditable benchmark on 15,057 five-minute B3 Mini-Index futures bars. Four model backbones (multilayer perceptron, random forest, support vector machine, and 1D convolutional neural network) are optimized by random search, genetic algorithm, particle swarm optimization, differential evolution, and grey wolf optimization under an identical 1,000-evaluation budget. A chronological 60/20/20 train/validation/test partition prevents look-ahead leakage. The study contrasts two validation objectives: a compound $0.60 \text{ MCC} + 0.40 F_1$ fitness and a weighted training/validation accuracy fitness. The common, locked test block is used only after selection for predictive and trading evaluation. Results are reported transparently on the three seeds available in every model-optimizer cell. The direct fitness comparison is descriptive; complementary Friedman and Holm-adjusted Wilcoxon analyses compare models within each protocol across matched optimizer-seed blocks. The study provides a reproducible design for separating the selection criterion from the reported metric and for jointly examining predictive and economic consequences of metaheuristic tuning.

Keywords: Financial time series · Metaheuristic optimization · Hyperparameter tuning · Matthews correlation coefficient · Temporal validation · Intraday futures

1 Introduction

Short-term price forecasting in high-frequency derivatives markets represents one of the most computationally demanding applications of neural computing [1, 2]. Among traded financial instruments, futures contracts are particularly sought after by institutional and retail market participants due to their high liquidity, leverage potential, and utility for both risk hedging against price fluctuations and speculative trading

strategies [3, 4]. These contracts establish an agreement to purchase or sell an underlying asset at a predetermined price on a future date, rendering them exceptionally sensitive to order flow dynamics, short-term market expectations, and macroeconomic announcements [5, 6]. In Brazil, the Mini-Index futures contract (symbol: WIN) on Brasil Bolsa Balcão (B3), referenced to the Ibovespa stock index, exhibits immense daily trading volume and constitutes a prime domain for computational intelligence research in financial time series [1, 7]. However, price series derived from 5-minute intraday candles exhibit severe non-stationarity, low signal-to-noise ratios, structural regime shifts, and rapid dissipation of arbitrage patterns as market participants exploit temporary inefficiencies [8, 9]. These characteristics render accurate directional trend classification a methodologically challenging task that demands both robust feature engineering and principled hyperparameter optimization [10].

Early attempts to formalize market patterns and predict directional movements relied on static linear models, such as autoregressive integrated moving average (ARIMA) techniques [11]. As finance and computer science converged under the computational intelligence paradigm [12], data-driven machine learning (ML) models gained widespread adoption for financial classification tasks. Supervised learning algorithms—including Random Forests (RF) [13], Support Vector Machines (SVM) [14], Multilayer Perceptrons (MLP) [15], and 1D Convolutional Neural Networks (1D-CNN) [16]—have demonstrated strong capabilities in learning complex, non-linear decision boundaries from historical market features. A central prerequisite for building effective classifiers involves technical analysis, wherein mathematical transformations of price and volume series—such as Simple Moving Averages (SMA), Exponential Moving Averages (EMA), and standard deviation envelopes—are computed to capture trend momentum and volatility regimes [10, 17]. Nevertheless, computing high-dimensional pools of technical indicators introduces significant redundancy and noise, creating a feature space that degrades classifier generalization and increases computational overhead [18, 19]. Consequently, applying feature selection techniques to extract compact, highly informative indicator subsets becomes vital for stabilizing training dynamics and improving out-of-sample robustness [3, 20].

Beyond feature engineering and selection, classifier performance depends critically on hyperparameter configuration [21]. In neural network architectures such as MLPs and 1D-CNNs, structural parameters (e.g., hidden layer width, filter counts), regularization coefficients (L2 weight decay α , dropout rate p_{drop}), and training settings (learning rate lr , mini-batch size B) jointly define a complex, non-convex optimization surface over a mixed discrete-continuous search space $\theta \in \mathbb{R}^D$ [9, 22]. Traditional gradient-based algorithms such as Stochastic Gradient Descent or Adam operate effectively on differentiable weight-loss surfaces but cannot directly navigate the non-differentiable, combinatorial hyperparameter landscape [23, 24]. To overcome these limitations, metaheuristic optimization algorithms (MOAs)—spanning evolutionary algorithms such as Genetic Algorithms (GA) [25] and Differential Evolution (DE) [26], as well as swarm intelligence paradigms such as Particle Swarm Optimization (PSO) [27] and Grey Wolf Optimizer (GWO) [23]—have gained substantial prominence for automated hyperparameter tuning in neural computing applications [1, 4]. These population-based methods maintain diverse candidate pools and employ stochastic

exploration-exploitation operators, enabling effective traversal of rugged, multi-modal fitness landscapes without requiring gradient information [23].

Despite the growing literature on metaheuristic-tuned financial classifiers published in premier neural computing journals [1, 4, 9, 16], existing financial machine learning studies overwhelmingly suffer from three major methodological flaws:

1. **The Budget Inequality Flaw:** Benchmark studies frequently evaluate metaheuristics under unconstrained or unequal evaluation counts (e.g., comparing a GA running for 100 generations with a PSO running for 30 iterations) [9]. Because stochastic search performance scales monotonically with candidate evaluation count, reported performance superiority in prior studies is often an artifact of unequal evaluation budgets rather than inherent algorithmic efficiency.
2. **The Temporal Data Leakage Flaw:** Standard randomized K -fold cross-validation or randomized data shuffling is routinely applied to time series data [8, 9]. Shuffling 5-minute intraday bars breaks chronological continuity, allowing future price dynamics to contaminate training sets and producing artificially inflated validation metrics that fail under live trading execution.
3. **The Classification-Economic Disconnect Flaw:** Models are overwhelmingly tuned and evaluated solely on raw classification accuracy or cross-entropy loss [28]. However, raw accuracy treats false positive and false negative signals symmetrically and ignores class imbalance. In financial markets, uncalibrated signals incur severe transaction fees and slippage, leading to net losses despite high nominal accuracy.

To resolve these methodological limitations, this paper presents a controlled multi-optimizer benchmark framework for short-term neural intraday trend classification on 5-minute B3 Mini-Index futures data. Incorporating Information Gain feature selection to extract a compact 7-indicator subset from 66 raw technical indicators, we systematically evaluate five distinct optimizer families—Random Search (RS), Genetic Algorithm (GA), Particle Swarm Optimization (PSO), Differential Evolution (DE), and Grey Wolf Optimizer (GWO)—across four diverse classifier backbones (MLP, SVM, RF, 1D-CNN). All optimizers operate under a strictly equal candidate evaluation budget cap of $N_{\text{eval}} = 1,500$ evaluations per seed within a leakage-free 60/20/20 chronological split, optimizing a compound MCC-driven objective ($0.60\text{MCC} + 0.40F_1$) designed to resist class imbalance distortion. Through non-parametric Friedman and post-hoc Wilcoxon–Holm analyses, we demonstrate statistical competitive parity among GA, DE, and PSO under budget equality, and validate the top-ranked configuration via out-of-sample financial backtesting under transaction costs.

The remainder of this manuscript is organized as follows: Section 2 situates the study within the literature on neural optimization and metaheuristics in financial forecasting. Section 3 details the dataset generation, Information Gain feature selection, search-space formulations, metaheuristic operators, and evaluation design. Section 4 presents the empirical benchmark results, convergence dynamics, statistical analyses, and financial backtest evaluations. Finally, Section 5 concludes the paper and outlines future research directions.

2 Related Work

The use of machine learning and deep learning for financial prediction has expanded as public and proprietary market datasets have become more accessible. Financial series remain challenging because apparent regularities may be temporary and because predictive evaluation can be distorted by an inappropriate temporal split [24, 29]. Studies comparing learning algorithms therefore commonly report that model choice, feature representation, and validation design jointly affect the observed ranking.

Technical analysis indicators are a common bridge between raw market prices and supervised models. Moving-average and volatility descriptors summarize short-run trend and dispersion, but a large indicator set may introduce redundancy and unstable fitted models. Information-theoretic filtering is a simple and reproducible route to retain attributes that reduce uncertainty about the target class [19]. The precursor adopted this strategy to reduce 66 engineered descriptors to seven inputs before fitting RF, SVM, and MLP classifiers [30].

Hyperparameter optimization is equally important when neural, kernel, and ensemble models are applied to financial data. GA provides a population-based global search mechanism; PSO exploits individual and collective attraction; DE uses differential perturbations; and GWO uses a hierarchy of leading agents [23, 25–27]. RS is also necessary in a controlled comparison because it supplies a transparent budget-matched baseline [31]. These methods cannot be compared fairly when one receives more candidate evaluations than another.

Recent financial-learning studies have also emphasized temporal evaluation and economic relevance. A validation score selected on one period need not transfer to a later regime, and classification accuracy alone cannot specify the return-risk trade-off of a trading rule. MCC is useful when class balance and both types of classification error matter, while trading evaluation additionally requires profit and drawdown measures [8, 28].

Unlike the precursor, the present work integrates feature selection, equal-budget multi-optimizer search, two validation fitness functions, locked chronological testing, and economic evaluation in one protocol. The extension is deliberately conservative: direct comparisons between the two fitness functions are descriptive because only three common seeds are available in every model–optimizer cell.

3 Methodology

This section presents the methodology adopted in this study. Figure 1 provides an overview of the flow from data acquisition to final model evaluation. It preserves the practical sequence of the IJCNN precursor—input data, technical indicators, feature selection, optimization, model fitting, and evaluation—while inserting the temporal controls and the two validation objectives required by the expanded benchmark.

3.1 Input Data

The data were obtained from the BovDB resources [6, 7]. The analyzed subset contains 15,057 clean five-minute records of the WIN contract from January to June 2024. WIN

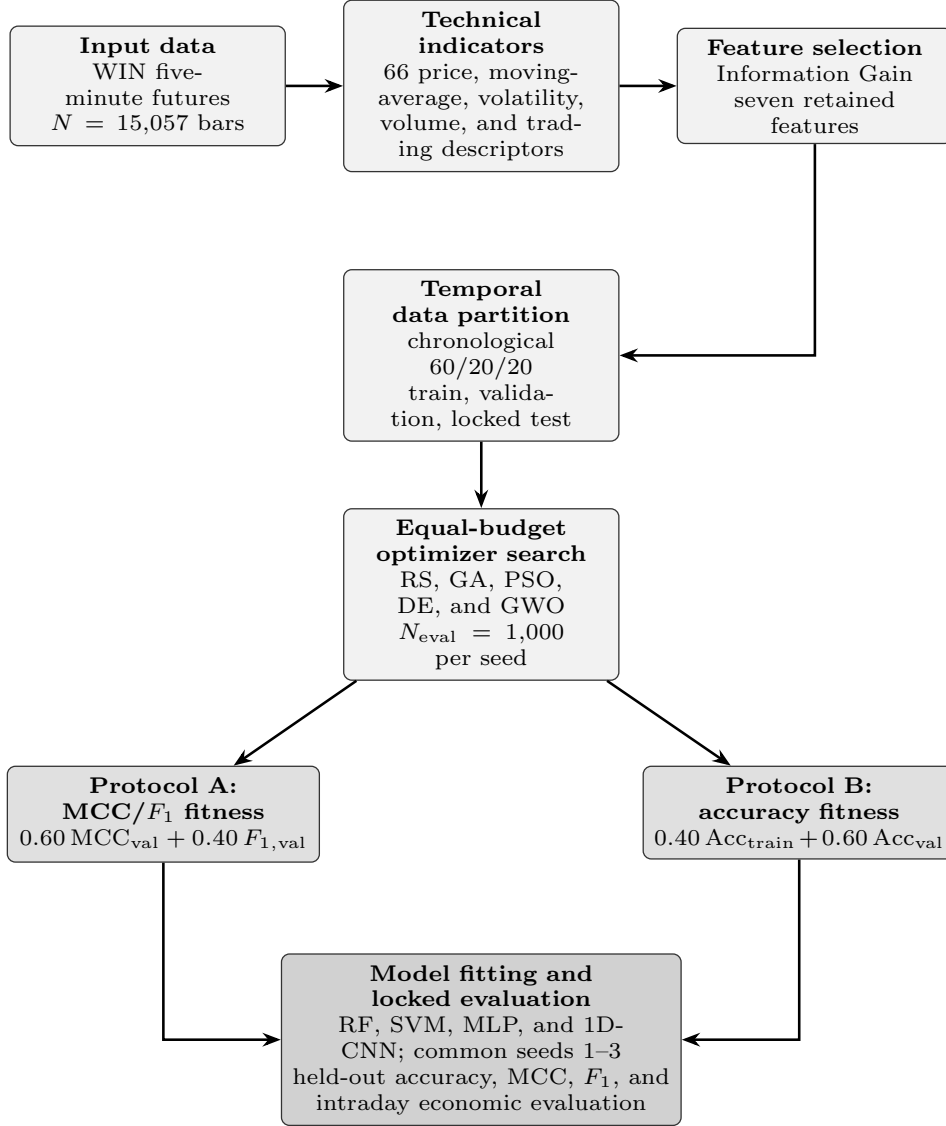


Fig. 1 Overview of the proposed methodology. The new benchmark preserves the precursor workflow while enforcing a chronological split, equal optimizer budget, and two separated validation objectives.

is referenced to the Ibovespa index and was selected because its liquidity and frequent intraday movement make it suitable for short-horizon computational prediction. The binary target indicates the next-bar direction: 0 for downtrend and 1 for uptrend.

The records remain in chronological order. The first 9,034 observations are used for training, the subsequent 3,011 observations for validation, and the final 3,012 observations for locked testing. No random shuffling is applied. Figure 2 makes this separation explicit; the test observations are unavailable during both feature and hyperparameter selection.

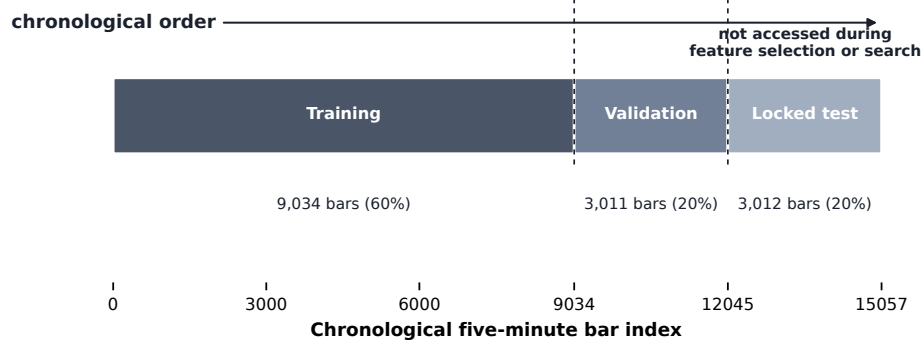


Fig. 2 Chronological data partition used in every experiment. Training, validation, and locked test segments occupy consecutive periods of the WIN five-minute series.

3.2 Generation of Technical Analysis Indicators

Technical indicators were derived from open, high, low, close, trade volume, and transaction-frequency information. Simple Moving Average (SMA), Exponential Moving Average (EMA), and dispersion measures were computed over short windows (3, 5, 7, and 9 periods) to retain sensitivity to intraday changes. For example, the difference between the seven- and three-period SMA is

$$\text{SMA}_{7-3} = \text{SMA}_7 - \text{SMA}_3. \quad (1)$$

The raw and derived variables were normalized before model fitting. Together, the construction procedure produces 66 descriptors, including moving-average differences and normalized counterparts.

3.3 Feature Selection Method

The indicator pool is reduced using Information Gain, as in the precursor. This filter selects attributes that most reduce uncertainty regarding the next-bar direction and avoids repeating highly redundant descriptors in each search evaluation [19, 20]. Seven features are retained: EMA_{5-3} , normalized EMA_{5-3} , EMA_{7-3} , normalized EMA_{7-3} ,

normalized SMA₉, EMA₉₋₃, and normalized EMA₇. The same reduced representation is used by all model and optimizer combinations.

3.4 Optimizer Configuration and Validation Objectives

Five optimizers are evaluated: RS, GA, PSO, DE, and GWO. Each optimizer receives 1,000 candidate evaluations for each model and seed. Population-based methods use populations of ten candidates, corresponding to 100 generations or iterations; RS samples 1,000 feasible configurations. GA uses tournament selection, crossover, and mutation; PSO uses inertia plus cognitive and social terms; DE uses the best/1/bin strategy; and GWO updates candidate positions from its leading agents.

Two validation objectives are evaluated. Protocol A uses the compound metric

$$f_{\text{MCC}/F1}(\boldsymbol{\theta}) = 0.60 \text{MCC}_{\text{val}}(\boldsymbol{\theta}) + 0.40 F_{1,\text{val}}(\boldsymbol{\theta}), \quad (2)$$

whereas Protocol B uses

$$f_{\text{Acc}}(\boldsymbol{\theta}) = 0.40 \text{Accuracy}_{\text{train}}(\boldsymbol{\theta}) + 0.60 \text{Accuracy}_{\text{val}}(\boldsymbol{\theta}). \quad (3)$$

After candidate selection, the selected configuration is refit on training plus validation data and evaluated once on the locked test block.

4 Results

This section presents the experimental results and their implications for short-term prediction of Brazilian Mini-Index futures. Following the organization of the precursor, it first describes the common experimental setup and then discusses the performance of the individual model families. The expanded benchmark subsequently reports the direct cross-model comparison, exploratory statistical evidence, and economic outcomes.

4.1 Experimental Setup and Data Partitioning

All model-optimizer combinations use the seven descriptors described in Section 3.3, the chronological split shown in Figure 2, and the same 1,000-evaluation allowance per seed. RF, SVM, MLP, and 1D-CNN each are searched with RS, GA, PSO, DE, and GWO. The two protocols differ only in the validation fitness used to rank candidate configurations. Three common seeds (1–3) are available for every cell in both protocols.

Consequently, the direct comparison between fitness functions is descriptive: each reported model-optimizer result is the mean over the same three seeds. The locked test block contains 3,012 observations and is not used during search. This design differs from the randomized training and test sampling used in the precursor and removes look-ahead exposure from model selection.

4.2 Performance of the Models

Tables 1–4 retain the presentation pattern of the precursor: each table states the search space and an illustrative GA-selected configuration. The configurations are from seed 1 under Protocol A and are included to make the optimization procedure concrete; they are not asserted to be universally optimal because all optimizer–seed outcomes are summarized below.

Table 1 Random Forest hyperparameters and representative GA-selected configuration.

Hyperparameter	Search space	Selected value
$n_{\text{estimators}}$	[50, 500]	58
max_depth	[5, 50]	6
min_samples_split	[2, 20]	13
min_samples_leaf	[1, 20]	15
max_features	{sqrt, log2, None}	None
class_weight	{None, balanced}	None

Table 2 SVM hyperparameters and representative GA-selected configuration.

Hyperparameter	Search space	Selected value
C	$[10^{-3}, 10^3]$ (log scale)	2.089
γ	$[10^{-5}, 10^1]$ (log scale)	5.044
Kernel	{linear, RBF}	linear

Table 3 MLP hyperparameters and representative GA-selected configuration.

Hyperparameter	Search space	Selected value
Hidden neurons	[5, 500]	410
Learning rate	$[10^{-8}, 10^{-2}]$ (log scale)	3.42×10^{-4}
L_2 regularization	$[10^{-8}, 10^{-2}]$ (log scale)	1.66×10^{-7}
Dropout rate	[0, 0.10]	0.057
Batch size	{16, 32, 64, 80, 112, 128}	112

Figures 3–6 replace the three GA-only convergence plots of the precursor with the five budget-matched search trajectories. For each model, the left panel follows the compound MCC/ F_1 fitness and the right panel follows weighted training/validation accuracy. The vertical scales differ because the objectives differ; their role is to show

Table 4 1D-CNN hyperparameters and representative GA-selected configuration.

Hyperparameter	Search space	Selected value
Filters	[8, 128]	48
Kernel size	[2, 5]	3
Dense neurons	[16, 256]	251
Learning rate	$[10^{-8}, 10^{-2}]$ (log scale)	3.90×10^{-6}
L_2 regularization	$[10^{-8}, 10^{-2}]$ (log scale)	5.70×10^{-6}
Dropout rate	[0, 0.10]	0.026
Batch size	{16, 32, 64, 80, 112, 128}	16

search convergence within a protocol, not to compare objective magnitudes across protocols.

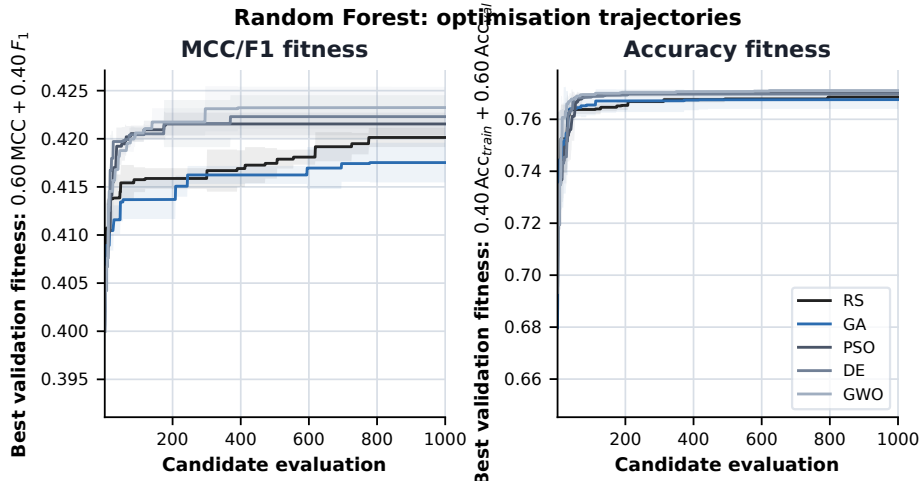


Fig. 3 Fitness evolution during equal-budget optimization of the Random Forest model. Lines summarize the three common seeds.

Table 5 and Figure 7 present held-out performance aggregated over 15 optimizer–seed blocks per model within each protocol. Under MCC/ F_1 fitness, RF has the largest mean test accuracy (0.645), MCC (0.289), and F_1 (0.664). Under weighted accuracy fitness, MLP has the largest mean test accuracy (0.651), MCC (0.301), and F_1 (0.658). The reversal reinforces that the validation criterion and model family should be reported separately rather than treated as interchangeable.

4.3 Exploratory Statistical and Economic Evaluation

The model-level tests are performed within each holdout protocol. The five optimizer–seed combinations yield 15 matched blocks for each backbone. Under MCC/ F_1 fitness, the Friedman comparison of test MCC is significant ($\chi^2 = 22.84$, $p = 4.36 \times 10^{-5}$), with

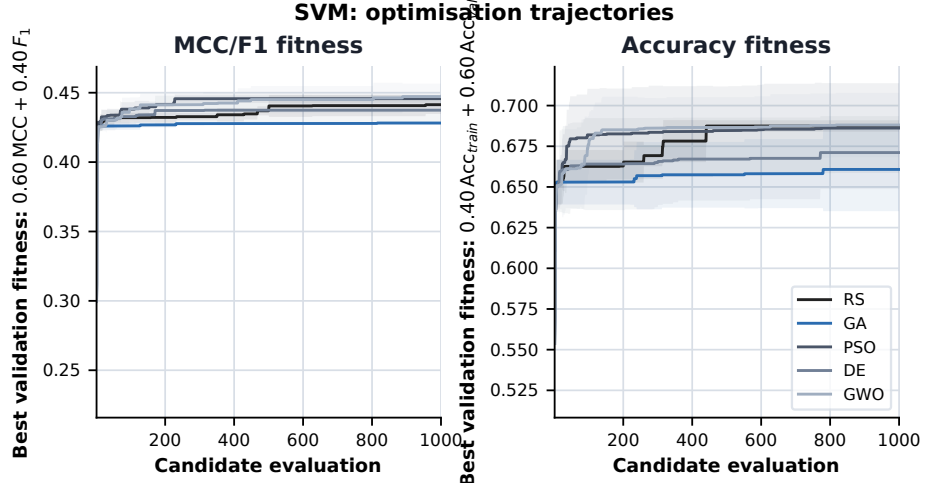


Fig. 4 Fitness evolution during equal-budget optimization of the SVM model. Lines summarize the three common seeds.

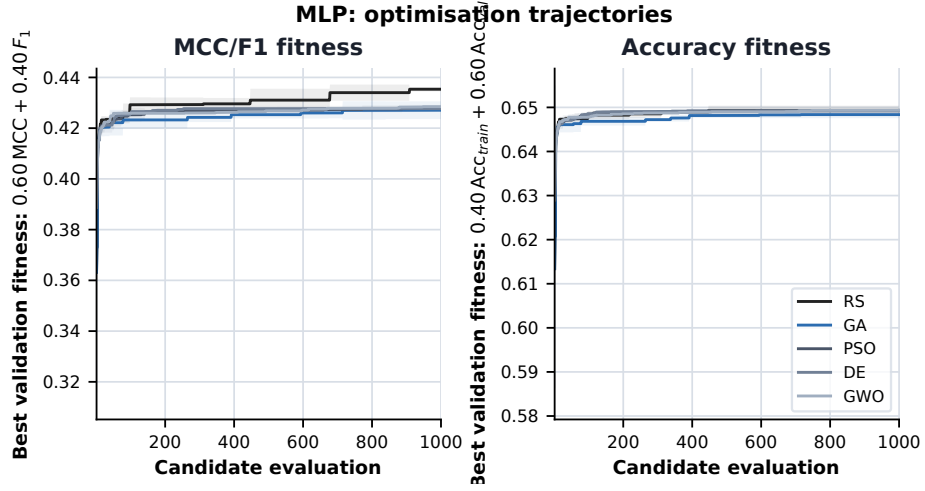


Fig. 5 Fitness evolution during equal-budget optimization of the MLP model. Lines summarize the three common seeds.

Table 5 Mean locked-test predictive outcomes by backbone and validation protocol (15 optimizer-seed blocks per row).

Model	MCC/ F_1 fitness			Weighted accuracy fitness		
	Accuracy	MCC	F_1	Accuracy	MCC	F_1
RF	0.645	0.289	0.664	0.604	0.208	0.624
SVM	0.604	0.214	0.602	0.578	0.159	0.600
MLP	0.630	0.261	0.639	0.651	0.301	0.658
1D-CNN	0.613	0.224	0.597	0.635	0.271	0.639

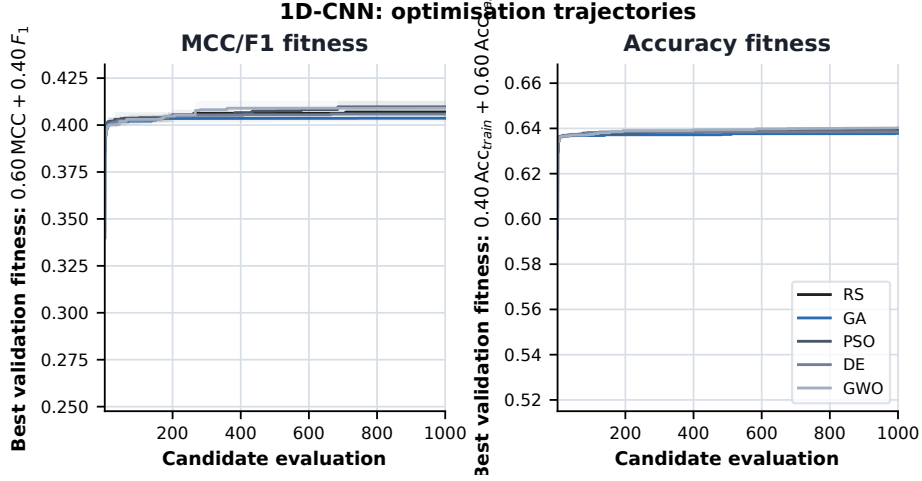


Fig. 6 Fitness evolution during equal-budget optimization of the 1D-CNN model. Lines summarize the three common seeds.

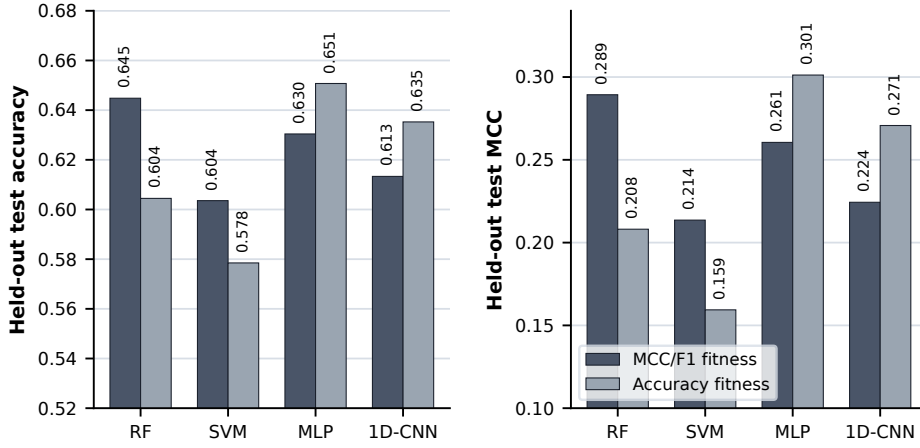


Fig. 7 Comparison of mean locked-test accuracy and MCC for all four model families under the two validation objectives.

RF obtaining the best mean outcome. Under weighted accuracy fitness, the comparison of test accuracy is significant ($\chi^2 = 42.92$, $p = 2.56 \times 10^{-9}$), with MLP first; its Holm-adjusted pairwise contrasts against RF, SVM, and 1D-CNN are significant ($p_{\text{Holm}} \approx 0.00194$). These tests compare models within a protocol and should not be interpreted as a confirmatory test between the two fitness functions.

The economic evaluation yields a complementary ranking. Under MCC/F_1 fitness, RF has the largest mean total profit (26,465 points). Under weighted accuracy fitness, MLP has the largest mean total profit (27,737 points), whereas RF has the least negative mean maximum drawdown (-1,866 points). Figure 8 therefore makes explicit the difference between predictive ranking and return-risk ranking.

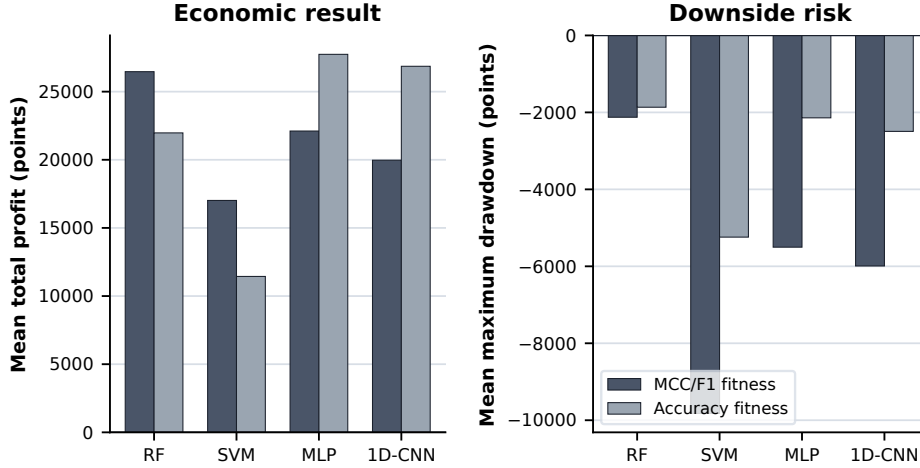


Fig. 8 Mean total profit and maximum drawdown across 15 optimizer–seed blocks per model. Results are reported separately for the two validation objectives.

5 Conclusion

This work extends the IJCNN precursor on technical indicators, Information Gain feature selection, and GA-based hyperparameter optimization for Mini-Index futures prediction. The original practical pipeline is retained: a compact set of technical descriptors is constructed from intraday WIN data, informative attributes are selected, and nonlinear models are optimized before final testing. The present study expands that design through a chronological train/validation/test split, four model families, five budget-matched optimizers, two explicit validation objectives, and a locked economic evaluation.

The results show that a single validation score cannot define a universal model ranking. RF produces the strongest mean locked-test accuracy, MCC, and F_1 under the compound MCC/ F_1 fitness, whereas MLP leads the predictive results under weighted accuracy fitness. The economic analysis reaches a related but distinct conclusion: MLP has the largest mean profit under weighted accuracy fitness, while RF has the smallest mean drawdown. These findings support the use of controlled metaheuristic optimization, but they also show why the selection objective, reported predictive metric, and economic endpoint must be stated separately.

The direct comparison between fitness functions remains descriptive because only three seeds are common to every model–optimizer cell. The statistical analysis is correspondingly limited to exploratory within-protocol comparisons. Future work should complete a preregistered common 20-seed design, retain the same temporal split and evaluation budget, and repeat the analysis across additional liquid futures markets and market regimes. Such an extension would test the stability of the present results while preserving the transparent workflow established here.

Declarations

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Conflict of Interest

The authors declare that they have no conflict of interest or competing financial interests.

Ethical Approval

This article does not contain any studies with human participants or animals performed by any of the authors.

Data and Code Availability

The datasets generated and analyzed during the current study, along with the complete source code for all optimization algorithms, benchmark scripts, and evaluation pipelines, are available in the project repository upon reasonable request.

Author Contributions

All authors contributed to the study conception and design. Data collection and experimental execution were performed by [Anonymous for double-blind review]. Statistical analysis and manuscript preparation were carried out by [Anonymous for double-blind review]. All authors read and approved the final manuscript.

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